MIL-C-27287/1 (USAF)
28 June 1960

MILITARY SPECIFICATION

CAPACITORS, FIXED, PLASTIC (NONMETALLIC CASES)

STYLE CTM-CHARACTERISTIC V

1. SCOPE

1.1 SCOPE.- This specification covers the requirements for Polyethylene terephthalate dielectric capacitors, fixed, style CTM for use in an operating temperature range of -55° to + 125°C.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on the date of invitation for bids, forms a part of this specification:

SPECIFICATION

Military

MIL-C-27287	Capacitors, Fixed, Plastic Dielectric, DC (Nonmetallic Cases), General Specification For

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 GENERAL.- The requirements, including materials and workmanship, of Specification MIL-C-27287 are applicable as requirements of this specification.

3.2 DETAIL.- The detail requirements shall be in accordance with figure 1 and table I.

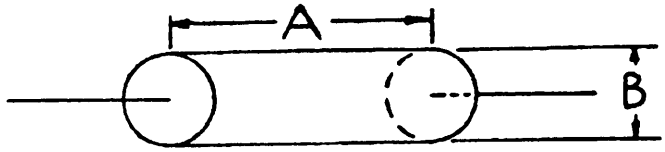


Figure 1

3.3 LEADS.- Leads shall be 2-5/8 maximum and 1-5/8 minimum in length, of tinned solid wire, 0.025 inch (No. 22 AWG) for cases 0.172 inch through 0.329 inch (incl) in diameter, and 0.032 inch (No. 20 AWG) for cases 0.360 inch and larger.

3.4 DIELECTRIC FILM.- Dielectric film utilized in the fabrication of capacitors shall be polyethylene terephthalate.

FSC 5910

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TABLE I

Type Designation	Rated Voltage in Volts DC 125°C	Rated Voltage in Volts DC *85°C	Nom. Capac. in uf	Case Dimensions B See Note 1	Case Dimensions A ± 1/16
CTM682VA-	50	100	.0068	.172	7/16
CTM822VA-	50	100	.0082	.172	7/16
CTM103VA-	50	100	.01	.172	7/16
CTM123VA-	50	100	.012	.172	11/16
CTM153VA-	50	100	.015	.172	11/16
CTM183VA-	50	100	.018	.188	11/16
CTM223VA-	50	100	.022	.188	11/16
CTM273VA-	50	100	.027	.203	11/16
CTM333VA-	50	100	.033	.203	11/16
CTM393VA-	50	100	.039	.250	11/16
CTM473VA-	50	100	.047	.250	11/16
CTM563VA-	50	100	.056	.250	13/16
CTM683VA-	50	100	.068	.250	13/16
CTM823VA-	50	100	.082	.297	13/16
CTM104VA-	50	100	.1	.297	13/16
CTM124VA-	50	100	.12	.360	13/16
CTM154VA-	50	100	.15	.360	13/16
CTM184VA-	50	100	.18	.329	1- 3/16
CTM224VA-	50	100	.22	.329	1- 3/16
CTM274VA-	50	100	.27	.390	1- 3/16
CTM334VA-	50	100	.33	.390	1- 3/16
CTM394VA-	50	100	.39	.485	1- 3/16
CTM474VA-	50	100	.47	.485	1- 3/16
CTM564VA-	50	100	.56	.500	1- 9/16
CTM684VA-	50	100	.68	.500	1- 9/16
CTM824VA-	50	100	.82	.548	1- 9/16
CTM105VA-	50	100	1.0	.548	1- 9/16
CTM332VB-	100	200	.0033	.172	7/16
CTM392VB-	100	200	.0039	.172	7/16
CTM472VB-	100	200	.0047	.172	7/16
CTM562VB-	100	200	.0056	.172	11/16
CTM682VB-	100	200	.0068	.172	11/16
CTM822VB-	100	200	.0082	.188	11/16
CTM103VB-	100	200	.01	.188	11/16
CTM123VB-	100	200	.012	.203	11/16
CTM153VB-	100	200	.015	.203	11/16
CTM183VB-	100	200	.018	.234	11/16
CTM223VB-	100	200	.022	.234	11/16
CTM273VB-	100	200	.027	.250	13/16
CTM333VB-	100	200	.033	.250	13/16
CTM393VB-	100	200	.039	.266	13/16
CTM473VB-	100	200	.047	.266	13/16

* 85°C rating for reference only

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TABLE I (continued)

Type Designation	Rated Voltage in Volts DC 125°C	Rated Voltage in Volts DC *85°C	Nom. Capac. in uf	Case Dimensions B See Note 1	Case Dimensions A ± 1/16
CTM563VB-	100	200	.056	.297	13/16
CTM683VB-	100	200	.068	.297	13/16
CTM823VB-	100	200	.082	.329	1- 1/16
CTM104VB-	100	200	.1	.329	1- 1/16
CTM124VB-	100	200	.12	.390	1- 1/16
CTM154VB-	100	200	.15	.390	1- 1/16
CTM184VB-	100	200	.18	.390	1- 3/16
CTM224VB-	100	200	.22	.390	1- 3/16
CTM274VB-	100	200	.27	.500	1- 3/16
CTM334VB-	100	200	.33	.500	1- 3/16
CTM394VB-	100	200	.39	.516	1- 9/16
CTM474VB-	100	200	.47	.516	1- 9/16
CTM272VC-	200	400	.0027	.188	11/16
CTM332VC-	200	400	.0033	.188	11/16
CTM392VC-	200	400	.0039	.203	11/16
CTM472VC-	200	400	.0047	.203	11/16
CTM562VC-	200	400	.0056	.250	11/16
CTM682VC-	200	400	.0068	.250	11/16
CTM822VC-	200	400	.0082	.250	13/16
CTM103VC-	200	400	.01	.250	13/16
CTM123VC-	200	400	.012	.266	13/16
CTM153VC-	200	400	.015	.266	13/16
CTM183VC-	200	400	.018	.313	13/16
CTM223VC-	200	400	.022	.313	13/16
CTM273VC-	200	400	.027	.360	13/16
CTM333VC-	200	400	.033	.360	13/16
CTM393VC-	200	400	.039	.360	1- 1/16
CTM473VC-	200	400	.047	.360	1- 1/16
CTM563VC-	200	400	.056	.390	1- 3/16
CTM683VC-	200	400	.068	.390	1- 3/16
CTM823VC-	200	400	.082	.485	1- 3/16
CTM104VC-	200	400	.1	.485	1- 3/16
CTM124VC-	200	400	.12	.500	1- 9/16
CTM154VC-	200	400	.15	.500	1- 9/16
CTM102VD-	300	600	.001	.172	11/16
CTM122VD-	300	600	.0012	.172	11/16
CTM152VD-	300	600	.0015	.172	11/16
CTM182VD-	300	600	.0018	.188	11/16
CTM222VD-	300	600	.0022	.188	11/16
CTM272VD-	300	600	.0027	.203	11/16
CTM332VD-	300	600	.0033	.203	11/16
CTM392VD-	300	600	.0039	.250	11/16
CTM472VD-	300	600	.0047	.250	11/16
CTM562VD-	300	600	.0056	.250	13/16
CTM682VD-	300	600	.0068	.250	13/16
CTM822VD-	300	600	.0082	.266	13/16

* 85°C rating for reference only

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Table I (Cont'd)

Type Designation	Rated Voltage in Volts DC 125°C	Rated Voltage in Volts DC *85°C	Nom. Capac. in mf	Case Dimensions B See Note 1	Case Dimensions A ± 1/16
CTM103VD-	300	600	.01	.266	13/16
CTM123VD-	300	600	.012	.313	13/16
CTM153VD-	300	600	.015	.313	13/16
CTM183VD-	300	600	.018	.360	13/16
CTM223VD-	300	600	.022	.360	13/16
CTM273VD-	300	600	.027	.360	1- 1/16
CTM333VD-	300	600	.033	.360	1- 1/16
CTM393VD-	300	600	.039	.390	1- 3/16
CTM473VD-	300	600	.047	.390	1- 3/16
CTM563VD-	300	600	.056	.485	1- 3/16
CTM683VD-	300	600	.068	.485	1- 3/16
CTM823VD-	300	600	.082	.500	1- 9/16
CTM104VD-	300	600	.1	.500	1- 9/16

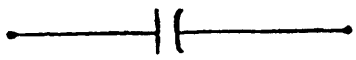
* 85°C rating for reference only

NOTE 1: Tolerances on dia.
± 0.031 below 0.250"
± 0.046 on 0.250 dia. thru 0.499
± 0.062 on 0.500 dia. and over

3.5 CONSTRUCTION.- Capacitors shall be extended foil construction.

3.6 TERMINAL.- The terminals shall be of the axial-lead type.

3.7 CIRCUIT.- The circuit shall be as shown in the diagram below:



3.8.1 CHARACTERISTIC.- These capacitors shall meet the following requirements for characteristic V.

3.8.1.1 INSULATION RESISTANCE.- The initial insulation resistance shall not be less than the applicable values shown in table II.

TABLE II.- Terminal to Terminal Insulation Resistance Measurements
For Characteristic V (See Figure 2)

Capacitance Rating	Minimum Insulation Resistance
25°C	
0 to 0.25 microfarads	100,000 megohms
Greater than 0.25 microfarads	25,000 megohms-microfarads
125°C	
0 to 0.067 microfarads	300 megohms
Greater than 0.067 microfarads	20 megohms-microfarads

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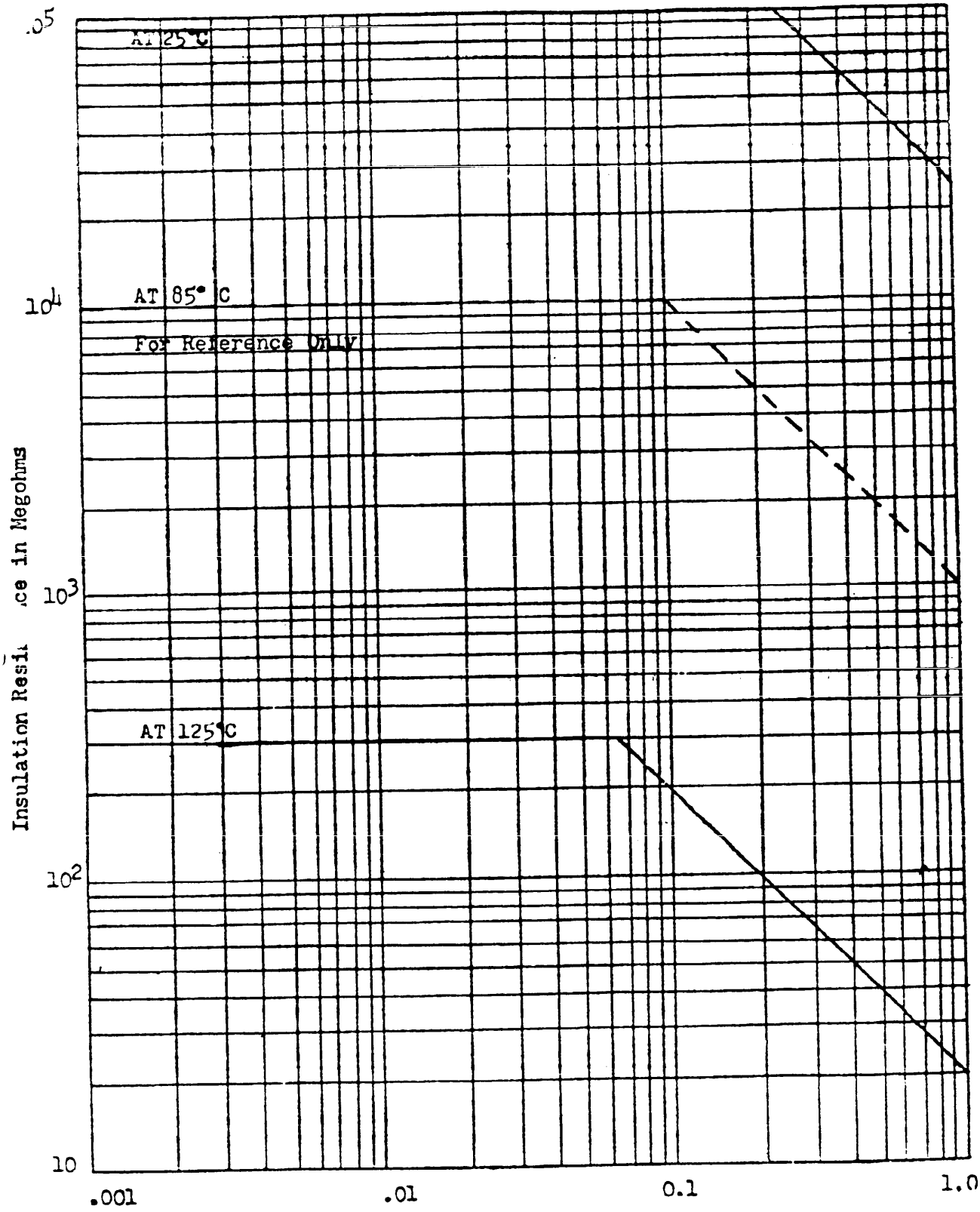


Figure 2 - Insulation Resistance Requirements

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3.8.1.2 CAPACITANCE CHANGE WITH TEMPERATURE.- Capacitance change at extreme temperatures shall not exceed the values specified in table III. For graphical representation of capacitance change throughout the temperature range see figure 3.

TABLE III.- Capacitance Change at Extreme Temperatures

Temperature Extremes	Capacitance Change in percent of 25°C value
-55°C	-10 %
+125°C	+20 %

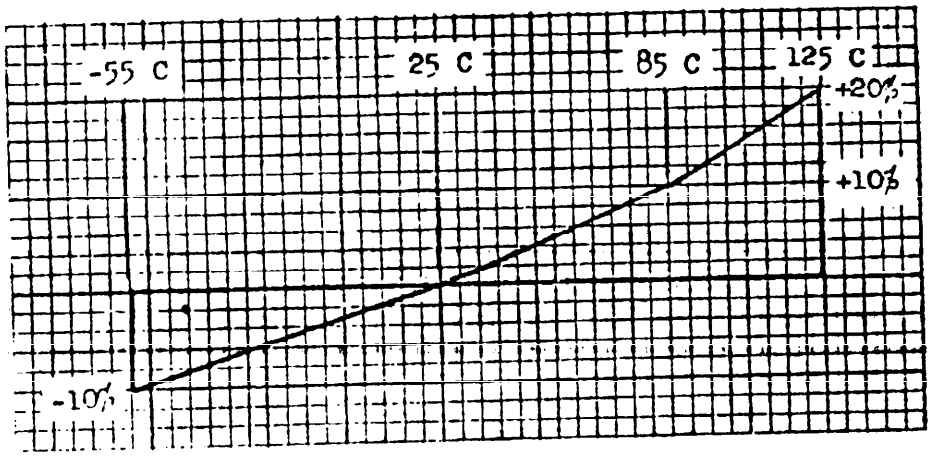


Figure 3
Graphical representation of capacitance change with temperature.

3.9 DIELECTRIC WITHSTANDING VOLTAGE.- The room temperature dielectric withstanding voltage applied shall be 400 percent of the rated dc voltage applied between terminals for a period of 1 minute. The following exceptions apply.

a. 350 percent rated dc voltage after temperature and immersion cycling and moisture-resistance tests.

b. For the contractor's inspection, the capacitors shall be subjected, at the option of the manufacturer, to the application of 500 percent of rated dc potential, for not less than 1 second, or 400 percent for not less than 15 seconds.

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4. QUALITY ASSURANCE PROVISIONS

4.1 GENERAL.- The sampling, inspection, and test procedures specified in Specification MIL-C-27287 are applicable as sampling, inspection, and test procedures of this specification.

5. PREPARATION FOR DELIVERY

5.1 GENERAL.- The capacitor shall be prepared for delivery in accordance with Specification MIL-C-27287.

6. NOTES

6.1 INTENDED USE.- Capacitors covered by this specification are intended for use in equipment having operating temperatures ranging from -55° to +125°C.

6.2 ORDERING DATA.- Ordering data shall be in accordance with Specification MIL-C-27287.

6.3 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is Commander, Wright Air Development Division, Attn: WWDPVE-2, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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